Main Checklist for Data_Project

Progress tracking for the DDR Data Logger project on Zybo Z7-20

Main Checklist

Stage	Stage Name	Status	Stage Output
0	GitHub + Project Structure	✓ **Done**	Repository, folders, README, docs, first commit
1	Data Protocol	✓ **Done**	docs/data_protocol.md with Header, Length, Data, Checksum
2	Data Generator + Testbench	✓ **Done**	rtl/data_generator.v, tb/tb_data_generator.v, simulation, waveform, report
3	Packet Builder FSM + Testbench	✓ **Done**	packet_builder_fsm.v + testbench + simulation
4	Sync FIFO + Testbench	✓ **Done**	sync_fifo.v + testbench + simulation
5	AXI-Stream Master + Testbench	✓ **Done**	axis_stream_master.v + testbench + simulation
6	Full RTL Top Simulation	☐ **Pending**	data_project_top.v + full testbench + simulation
7	Vivado Block Design + AXI DMA + DDR	☐ **Pending**	Connected Block Design ready to run
8	Vitis C Verification	☐ **Pending**	C software that starts DMA and verifies DDR data
9	Async FIFO + CDC + Testbench	☐ **Pending**	async_fifo.v + testbench with two clocks
10	Full Integration with CDC + DMA + DDR	☐ **Pending**	Full system with two clock domains and DDR
11	ILA + Vivado Reports	☐ **Pending**	ILA, timing report, utilization report
12	Python CSV / Graph	☐ **Pending**	CSV, graph, and documentation
13	UART or SPI Extension	☐ **Pending**	Real data source replacing Data Generator
14	Portfolio Polish	☐ **Pending**	GitHub, README, images, explanations, and reports polish

> Current status: Stages 0-5 are complete. The next stage is Stage 6 - Full RTL Top Simulation.

Detailed Checklist for Work Tracking

Stage 0 - GitHub + Project Structure

✓ Done

- ☑ Created Data_Project folder
- ☑ Created GitHub repository
- ☑ Created README.md
- ☑ Created organized folder structure
- ☑ Created work plan document
- ☑ Added block diagram
- ☑ Completed first commit
- ☑ Completed push to GitHub

Stage 1 - Data Protocol

✓ Done

- ☑ Created docs/data_protocol.md
- ☑ Defined DATA_WIDTH = 32
- ☑ Defined Header
- ☑ Defined Length
- ☑ Defined Data sequence
- ☑ Defined Checksum
- ☑ Defined Expected Packet
- ☑ Defined PASS / FAIL Criteria
- ☑ Completed commit
- ☑ Completed push

Stage 2 - Data Generator + Testbench

✓ Done

- ☑ Defined block goal
- ☑ Defined inputs and outputs
- ☑ Wrote rtl/data_generator.v
- ☑ Wrote tb/tb_data_generator.v
- ☑ Ran Behavioral Simulation
- ☑ Verified expected sequence: 1, 7, 8, 3, 6, 3
- ☑ Saved waveform screenshot: images/stage2_data_generator_waveform.png
- ☑ Created simulation report: reports/stage2_data_generator_simulation.md
- ☑ Completed commit
- ☑ Completed push

Stage 3 - Packet Builder FSM + Testbench

✓ Done

- ☑ Define block goal
- ☑ Define FSM states
- ☑ Write rtl/packet_builder_fsm.v
- ☑ Write tb/tb_packet_builder_fsm.v
- ☑ Run simulation
- ☑ Verify Header / Length / Data / Checksum
- ☑ Complete commit
- ☑ Complete push

Stage 4 - Sync FIFO + Testbench

✓ Done

- ☑ Define block goal
- ☑ Write rtl/sync_fifo.v
- ☑ Write tb/tb_sync_fifo.v
- ☑ Verify write/read
- ☑ Verify full/empty
- ☑ Verify overflow/underflow
- ☑ Complete commit
- ☑ Complete push

Detailed Checklist for Work Tracking

Stage 5 - AXI-Stream Master + Testbench

✓ Done

- [x] Define block goal
- [x] Write rtl/axis_stream_master.v
- [x] Write tb/tb_axis_stream_master.v
- [x] Verify tvalid / tready / tdata / tlast
- [x] Verify backpressure
- [x] Complete commit
- [x] Complete push

Stage 6 - Full RTL Top Simulation

☐ Pending

- [ ] Write rtl/data_project_top.v
- [ ] Write tb/tb_data_project_top.v
- [ ] Connect all blocks
- [ ] Run full simulation
- [ ] Save result
- [ ] Complete commit
- [ ] Complete push

Stage 7 - Vivado Block Design + AXI DMA + DDR

☐ Pending

- [ ] Create Block Design
- [ ] Add Zynq PS
- [ ] Add AXI DMA
- [ ] Add Interconnect / SmartConnect
- [ ] Connect Custom RTL
- [ ] Connect DDR
- [ ] Pass design validate
- [ ] Complete commit
- [ ] Complete push

Stage 8 - Vitis C Verification

☐ Pending

- [ ] Create Vitis project
- [ ] Write DMA initialization code
- [ ] Define DDR buffer address
- [ ] Start DMA transfer
- [ ] Read data from DDR
- [ ] Verify Header / Length / Data / Checksum
- [ ] Complete commit
- [ ] Complete push

Stage 9 - Async FIFO + CDC + Testbench

☐ Pending

- [ ] Write rtl/async_fifo.v
- [ ] Write tb/tb_async_fifo.v
- [ ] Verify two clocks
- [ ] Verify correct data crossing
- [ ] Complete commit
- [ ] Complete push

Stage 10 - Full Integration with CDC + DMA + DDR

☐ Pending

- [ ] Integrate Async FIFO into the system
- [ ] Update full integration
- [ ] Verify two clock domains
- [ ] Verify DDR write
- [ ] Complete commit
- [ ] Complete push

Detailed Checklist for Work Tracking

Stage 11 - ILA + Vivado Reports

☐ Pending

- ☐ Add ILA
- ☐ Verify key signals
- ☐ Save timing report
- ☐ Save utilization report
- ☐ Save screenshots
- ☐ Complete commit
- ☐ Complete push

Stage 12 - Python CSV / Graph

☐ Pending

- ☐ Create CSV file
- ☐ Write Python script
- ☐ Create graph
- ☐ Save screenshot / output
- ☐ Complete commit
- ☐ Complete push

Stage 13 - UART or SPI Extension

☐ Pending

- ☐ Choose UART or SPI
- ☐ Write communication block
- ☐ Run verification
- ☐ Receive real data through the system
- ☐ Complete commit
- ☐ Complete push

Stage 14 - Portfolio Polish

☐ Pending

- ☐ Update README
- ☐ Add images
- ☐ Add waveforms
- ☐ Add reports
- ☐ Add explanations for each block
- ☐ Add Future Work / Lessons Learned
- ☐ Prepare project for presentation